

Amritpal Kaur Dhillon

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PROFESSIONAL EXPERIENCE

REDYNOX, Remote Cybersecurity Intern | 6 Months

- **Executed** vulnerability assessments and penetration testing for mobile and web applications to identify critical security flaws.
- **Analysed** cybersecurity breach case studies to develop incident response strategies and improve organizational threat intelligence.
- **Implemented** network security protocols including CCNA-level switching and routing to secure enterprise infrastructure.

CODETECH IT SOLUTIONS, Remote AI/ML Intern | 7 Months

- **Designed** interactive data visualizations using IBM tools to translate complex datasets into actionable business insights.
- **Performed** exploratory data analysis (EDA) and machine learning modelling to identify patterns in large-scale datasets.
- **Streamlined** data engineering workflows by applying foundational systems analysis and cloud computing principles.

DEVELOPERS ARENA, Remote Data Science Intern | 6 Months

- **Engineered** predictive models to classify stress levels in employees and students, utilizing Random Forest and Neural Networks.
- **Optimized** university enrolment processes by developing an AI-powered lead scoring system to prioritize high-potential candidates.
- **Applied** K-means clustering algorithms to crime datasets to identify spatial-temporal hotspots and forecast high-risk areas.

CODETECH IT SOLUTIONS, Remote Full Stack Intern (Python) | 6 Months

- **Developed** responsive web applications using Python-based frameworks and integrated front-end components for seamless user experiences.

- **Architected** and managed database schemas to ensure efficient data retrieval and storage for high-traffic web platforms.
- **Optimized** application performance and scalability by implementing best practices in full-stack development and modular coding.

CODEC TECHNOLOGIES, Remote Python Developer Intern | 6 Months

- **Engineered** robust Python scripts to automate repetitive data processing tasks, increasing overall operational efficiency.
- **Collaborated** on the design and implementation of backend APIs to facilitate secure data communication between system modules.
- **Resolved** complex software bugs through rigorous debugging and unit testing, ensuring the delivery of high-quality, stable code.

INFOWIZ, Chandigarh, India

- **Industrial Trainee (Python) | 3 Months**
- **Completed** intensive industrial training focused on advanced Python programming and real-world software development lifecycles.
- **Built** scalable applications by applying object-oriented programming (OOP) principles and efficient algorithm design.
- **Leveraged** Python libraries to conduct data manipulation and analysis, preparing datasets for various technical applications.

RESEARCH PUBLICATIONS

- **Detection of Malicious Social Bots Using Learning Automata with URL Features in Twitter Network; *National Conference on Recent Advances in Science, Engineering, Technology and Management*, Ambala, India (Feb 2026). Paper ID: WR-NCR-AMBL-160226-14630.**
- **Drug Recommendation System Based on Sentiment Analysis of Drug Reviews Using Machine Learning, *National Conference on Recent Advances in Science, Engineering, Technology and Management*, Jaipur, India (March 2026). Paper ID: - WR-NCR-JPR-090326-14891**
- **A Sentiment Polarity Categorization Technique for Online Product Reviews; *International Conference on Research in Science, Engineering and Technology (ICRSET)*, Mumbai, India (April, 2026). Paper ID: WR-RSET-MBAI-030426-14859**

PROJECTS

- **Robotic Rubik's Cube Solver:** Built an autonomous robotic system integrating computer vision and Kociemba's Algorithm for real-time cube manipulation.
- **Fraud App Detection:** Developed a sentiment analysis tool using NLP to detect fraudulent mobile applications based on user review patterns.
- **Arduino Home Automation:** Designed a Bluetooth-controlled smart home system using Arduino Uno to enable wireless appliance management.
- **RFID Attendance System:** Engineered an automated attendance logger with real-time SMS alerts via GSM modem to enhance institutional transparency.
- **Detection of Malicious Social Bots:** Developed an intelligent system using Learning Automata and URL-based behavior analysis to identify phishing bots in Twitter networks, resulting in a research publication.
- **Knee Osteoarthritis Detection:** Engineered a deep learning model to detect and predict the severity of Knee Osteoarthritis from medical imaging to assist in early clinical diagnosis, resulting in journal publication.
- **Stress Detection System:** Applied Machine Learning models (Decision Trees, SVM, Random Forest) to classify stress levels in IT employees and students based on physiological and behavioral data.
- **Alcohol Detector:** Created a safety-focused embedded system utilizing an MQ-3 gas sensor and buzzer indicator to provide real-time alerts for high alcohol concentration levels.

EDUCATION

Manipal University Jaipur, Jaipur, India | *Expected 2027* (M. TECH)

- **Master of Technology (M. Tech) in Computer Science (AI/ML) | 2025-2027**
Chandigarh University, Mohali, India | 2022-2025 (B. TECH)
- **Bachelor of Technology (B. Tech) in Computer Science | Grade: 7.27/10**
- **Chandigarh Polytechnic College (PSBTE BOARD), Mohali, India | 2022 (DIPLOMA)**
- **Diploma in Computer Science and Engineering | Grade: 79.05 (2019-2022)**
- **Awards: MOOC Excellence Award 2025 for outstanding performance on Coursera.**

- **Holy Heart Presidency School, Amritsar, Punjab, India | 2016-2017**
 - **10th CLASS (ICSE BOARD)**
 - **Sidana International School, Amritsar, Punjab, India | 2017-2018**
 - **11th CLASS (CBSE BOARD) (SUPER MEDICAL)**
 - **Sidana International School, Amritsar, Punjab, India | 2018-2019**
 - **12th CLASS (CBSE BOARD) (SUPER MEDICAL)**
 - **ATTENDED WISDOM COACHING INSTITUTE AND PREPARED FOR NEET AND JEE MAINS**
 - **NEET AND JEE MAINS EXAM PREPARATION (2017-2019)**
 - **ATTENDED COACHING CLASSES ALONG WITH BTECH FROM RAJ MALHOTRA IAS INSTITUTE, CHANDIGARH**
 - **UPSC PREPARATION (2022-2025)**
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SKILLS & CERTIFICATIONS

- **Skills:** Python, Java, React.js, Power BI, SQL, Azure/AWS, Machine Learning, Penetration Testing.
- **Certifications:** AWS Certified Cloud Practitioner (2025), CCNA: Enterprise Networking, Security, and Automation, IBM Data Science Orientation.
- More Certifications from Coursera, Udemy, Great Learning, EC-Council, UpGrad, TuteDude
- **Languages:** English, Hindi, Punjabi, German, Spanish.